7. Transaction Count by Type

Aim

To write a COBOL program to count deposits and withdrawals separately and calculate their totals, reporting `INVALID BATCH` if a transaction type is invalid.

Algorithm

1. Start and read `N`.
2. Initialize deposit and withdrawal counts and totals to zero.
3. Read each transaction type and amount.
4. If the type is `D`, increment the deposit count and total.
5. If the type is `W`, increment the withdrawal count and total.
6. Otherwise, mark the batch as invalid.
7. After processing all records, display `INVALID BATCH` if any type is invalid; otherwise, display the counts and totals.
8. Stop.

Program

```
IDENTIFICATION DIVISION.

PROGRAM-ID. TRANS-COUNT.

DATA DIVISION.

WORKING-STORAGE SECTION.

01 N PIC 9(4).

01 I PIC 9(4).

01 T PIC X.
```

```
01 AMT PIC 9(7).
01 DC PIC 9(4) VALUE 0.
01 WC PIC 9(4) VALUE 0.
01 DT PIC 9(9) VALUE 0.
01 WT PIC 9(9) VALUE 0.
01 INVALID-FLAG PIC X VALUE 'N'.

PROCEDURE DIVISION.
   ACCEPT N
   PERFORM VARYING I FROM 1 BY 1 UNTIL I > N
      ACCEPT T
      ACCEPT AMT
      EVALUATE T
         WHEN 'D'
            ADD 1 TO DC
            ADD AMT TO DT
         WHEN 'W'
            ADD 1 TO WC
            ADD AMT TO WT
         WHEN OTHER
            MOVE 'Y' TO INVALID-FLAG
      END-EVALUATE
   END-PERFORM

   IF INVALID-FLAG = 'Y'
      DISPLAY "INVALID BATCH"
```

```
    ELSE
        DISPLAY "Deposit count: " DC
        DISPLAY "Deposit total: " DT
        DISPLAY "Withdrawal count: " WC
        DISPLAY "Withdrawal total: " WT
    END-IF
    STOP RUN.
```

Sample Input

3

D

100

W

30

D

50

Output

